

DARWIN'S ONE LONG ARGUMENT AND ESCOHOTADO'S THREE LONG VOLUMES:

Historical Inference, Institutional Vestiges, and the Methodological Convergence of Evolutionary Biology and Anti-Commercial Genealogy

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Abstract

Stephen Jay Gould argued that Darwin's self-described "one long argument" in *On the Origin of Species* (1859) was not merely a proof of evolution but a systematic methodology for historical inference in science. Gould identified four methods Darwin deployed: uniformity (extrapolation from observable processes), sequencing (ordering coexisting configurations as temporal phases), consilience (convergence of evidence from independent domains), and discordance (reading imperfections as vestiges of historical descent).

This paper demonstrates that Antonio Escohotado, in his three-volume *Los enemigos del comercio* (2008, 2013, 2017), independently employed three of Darwin's four methods to document the persistence of anti-commercial ideology across 2,600 years of human history.

Escohotado extrapolated from well-documented episodes to less visible ones, ordered societies as phases of a recurring anti-commercial cycle, and accumulated evidence from domains so diverse that case-by-case explanation became less parsimonious than a unifying pattern. The method he did not employ, discordance, is precisely the one that Extended Phenotype Theory (EPT) and the concept of the institutional palimpsest (Dawkins 2024) supply: reading institutional imperfections, vestigial norms, and regulatory hysteresis as evidence of evolutionary descent rather than design failure.

The paper connects this methodological analysis to the Smith-Darwin / Dawkins-Hayek genealogical convergence, arguing that Escohotado's empirical program and the EPT framework are complementary branches of a single intellectual tradition rooted in Adam Smith's insight that complex order emerges without a designer. Three falsifiable predictions are derived from the integrated framework.

Keywords: Extended Phenotype Theory, historical inference, Gould, Darwin, Escohotado, institutional vestiges, consilience, palimpsest, anti-commercial ideology, evolutionary game theory, Adam Smith, regulatory hysteresis

I. INTRODUCTION: TWO LONG ARGUMENTS SEPARATED BY 150 YEARS

Charles Darwin described *On the Origin of Species* as "one long argument" (Darwin 1859, p. 459). The phrase is typically read as false modesty or rhetorical understatement. Gould (2002) argued it was neither. Darwin meant exactly what he said: the book was a single, sustained exercise in historical inference, designed to argue that rigorous science could be conducted about events no one had witnessed, using evidence preserved in the present to reconstruct causes operating in the past.

Antonio Escohotado's *Los enemigos del comercio* (2008, 2013, 2017) is, in a structurally precise sense, three long arguments. Three volumes, thousands of pages, hundreds of primary sources in half a dozen languages, tracking a single phenomenon across 2,600 years: the recurrent emergence, mutation, adaptation, and survival of the idea that private property is theft and commerce its instrument. From Sparta to the Soviet Union, from the Essenes to Marx, from the Jacobins to Castro, Escohotado documented with entomological precision how this idea resurfaces after every defeat, adopts new vocabulary while preserving identical structure, and persists despite the consistent impoverishment of every society that implements it.

The structural parallel between Darwin's project and Escohotado's has not been noted in the literature. This is perhaps unsurprising: Escohotado was a lawyer, philosopher and historian of pharmacology, not a biologist, and his methodological toolkit was genealogical rather than formal. He did not cite Darwin, Dawkins, or evolutionary game theory. He did not need to; his project was different. But the methods he deployed to establish his case bear a striking resemblance to the methods Darwin deployed to establish his, as analyzed by Gould in *The Structure of Evolutionary Theory* (2002).

This paper makes three claims.

First, Escohotado employed three of the four methods of historical inference that Gould identified in Darwin's argumentative strategy: uniformity, sequencing, and consilience.

Second, the method he did not employ, discordance (the reading of imperfections as vestiges of historical descent), is precisely the method that Extended Phenotype Theory supplies through the concept of the institutional palimpsest and the mechanism of regulatory hysteresis.

Third, this methodological convergence is not accidental but reflects the deeper genealogical unity of Darwin's evolutionary biology and the Austrian tradition of spontaneous order analysis, both of which descend from Adam Smith's foundational insight that complex functional order can emerge without a designer.

The Dawkins-Hayek convergence, which I have developed in prior work (Lerer 2025a), provides the theoretical trunk from which both Escohotado's empirical genealogy and the EPT framework branch.

The argument proceeds as follows. Section II reconstructs Gould's analysis of Darwin's four methods. Section III examines Escobedo's deployment of three of the four, with attention to the partial independence problem that distinguishes his consilience from Darwin's. Section IV develops the case that institutional discordance, the fourth method, is what EPT adds to Escobedo's program, connecting the palimpsest concept from Dawkins (2024) with regulatory hysteresis. Section V traces the Smith-Darwin / Dawkins-Hayek genealogical trunk. Section VI specifies what the integrated framework predicts. Section VII discusses limitations. Section VIII concludes.

II. GOULD'S FOUR METHODS OF HISTORICAL INFERENCE IN DARWIN

Gould (2002, Chapter 2) argued that Darwin faced an epistemological challenge as formidable as the biological one: how to establish that historical science could achieve the rigor of experimental science. Since past processes cannot be observed empirically in real time, Darwin had to demonstrate how to infer causes from results, giving natural history the status of exact science. To overcome an intellectual environment dominated by fixism and creationism, Darwin designed a continuum of four inferential methods, ordered by the availability of information.

A. Uniformity: Extrapolation from Observable Processes

The first method takes direct observation of small-scale changes in the present and extrapolates them as mechanisms for large-scale changes over geological time. Darwin's paradigmatic example was artificial selection: pigeon breeders and horticulturists demonstrably modify organisms across generations through selective breeding. If human selection can produce the fantail pigeon from the rock dove in centuries, natural selection operating over millions of years can produce the diversity of life from common ancestors.

The method requires an assumption: that the processes observable in the present operated in the past with sufficient regularity to account for the results we see. This is not naive uniformitarianism (the claim that rates and intensities were constant); it is methodological uniformitarianism (the claim that the same types of causes operated). Darwin was explicit about the distinction, as Gould (2002) documented.

B. Sequencing: Ordering Coexisting Configurations as Temporal Phases

When direct observation of transitions is unavailable, Darwin ordered diverse configurations existing simultaneously as if they were stages of a single temporal process. His paradigmatic application was the argument against the objection that incipient stages of complex organs (the eye being the canonical example) could not have been functional. By presenting a gradation of existing eye structures, from simple light-sensitive cells in flatworms to the camera eye of vertebrates, all coexisting in the present, Darwin demonstrated that each stage was functional for

its bearer. The sequence of existing structures served as a proxy for the temporal sequence of evolutionary stages that could not be directly observed.

The method does not require that the existing structures represent an actual ancestral lineage. The flatworm eye is not the ancestor of the vertebrate eye. The argument is that the existence of functional intermediates at every level of complexity refutes the claim that intermediate stages would have been useless. The ordering is logical, not genealogical, though genealogical connections may underlie some sequences.

C. Consilience: Convergence from Independent Domains

William Whewell (1840) coined the term "consilience of inductions" to describe the situation in which evidence from independent domains converges on a single explanation. Darwin made consilience the backbone of his argument. The geographical distribution of species, the fossil record, comparative anatomy, embryology, vestigial organs, the classification of organisms into nested hierarchies: each domain independently pointed toward common descent with modification. No single domain was decisive; the power of the argument resided in the convergence.

Gould (2002) emphasized that this was Darwin's most deliberate tactical weapon against the theological establishment. William Paley's natural theology could accommodate any single anomaly by invoking the inscrutable purposes of the Creator. What it could not accommodate was the relentless accumulation of anomalies from unrelated fields, all pointing in the same direction. The convergence made case-by-case creationist explanations less parsimonious than the unifying evolutionary explanation. The cost of maintaining the creationist position rose with every additional domain Darwin recruited into the argument.

The independence of domains is crucial to the method's force. Biogeography has no causal connection to embryology; the fossil record has no causal connection to the classification of living species. When causally unconnected lines of evidence converge on the same conclusion, the probability that the convergence is coincidental diminishes with each additional line. This is what made Darwin's consilience devastating and what distinguishes it from mere accumulation of examples within a single domain.

D. Discordance: Reading Imperfections as Vestiges of Descent

The fourth method operates when information is minimal. Darwin inferred evolutionary history from a single object by analyzing its peculiarities, imperfections, or "extravagances" that did not fit its current mode of life. Rudimentary organs, the paradigmatic example: fetal teeth in baleen whales that never erupt, the vestigial eyes of cave-dwelling fish, the tiny wings of flightless beetles. These structures make no sense as optimal design for current function. They make perfect sense as inheritances from ancestors where they were functional.

Gould (2002) identified discordance as Darwin's most radical subversion of Paley's argument. Darwin accepted Paley's empirical observation that organisms exhibit exquisite adaptation. But he demonstrated that alongside the adaptations lie imperfections, vestiges, and jury-rigged solutions that no competent designer would have produced. The coexistence of adaptation and imperfection was inexplicable under design theory but predicted by descent with modification: selection improves what it finds, but it cannot start from scratch, and the traces of history accumulate in every lineage.

Dawkins (2024) extended this insight in *The Genetic Book of the Dead*. Every organism, Dawkins argued, is a palimpsest: a document in which earlier writings have been partially overwritten by later ones, but never fully erased. The genome carries the record of ancestral environments, readable through the morphological, physiological, and behavioral features that no longer serve their original function but persist because the cost of complete removal exceeds the cost of toleration. The human lumbar spine protests vertical posture because it was built for quadrupedal locomotion over hundreds of millions of years and superficially overwritten for bipedalism in a mere six million. The palimpsest is the biological instantiation of discordance as a method: read the imperfection, infer the history.

III. ESCOHOTADO'S METHODOLOGICAL ARSENAL: THREE OF FOUR

Escohotado did not cite Gould, Darwin, or evolutionary biology. His intellectual lineage was different: Hegel, the Frankfurt School (as target of critique), classical liberalism, and the history of pharmacology. His method was genealogical in the Nietzschean sense: tracing the descent of an idea through its successive incarnations, documenting the mutations it underwent, the environments it colonized, and the selective pressures that shaped its evolution. Yet the inferential methods he deployed are structurally isomorphic to three of Darwin's four.

A. Uniformity: Extrapolation from Documented to Underdocumented Periods

Escohotado's most thoroughly documented episodes are those with abundant primary sources: the French Revolution and the Terror (September 1793 to July 1794), the Russian Revolution and Soviet collectivization (1917 to 1933), the Cuban experiment (1959 onward). In each case, he reconstructed the anti-commercial mechanism with granular precision: the ideological justification, the institutional apparatus, the economic consequences, the political dynamics of radicalization.

For earlier periods where documentation is sparser, the Essenes, early Christian communities, medieval heretical movements, Escohotado extrapolated from the well-documented mechanism. If the same structure (communal property, prohibition of commerce, stigmatization of merchants, concentration of economic power in a priestly or political elite) is observable in detail where sources are rich, it is reasonable to infer that the same structure operated where sources are

fragmentary but the available evidence points in the same direction. This is precisely Darwin's uniformity method: take an observable process and extend it to domains where direct observation is impossible.

Consider a specific instance. Escobedo's account of the Jacobin Terror is based on abundant documentary evidence: the decrees of the Committee of Public Safety, the records of the Revolutionary Tribunal, Robespierre's speeches, Saint-Just's reports, contemporary memoirs, and economic data on prices, shortages, and black market activity. From this material he reconstructs a precise mechanism: the stigmatization of merchants as "accapareurs" (hoarders), the imposition of maximum prices (the Loi du Maximum General of September 1793), the creation of a police apparatus to enforce price controls, the emergence of a black market that the controls themselves generated, and the radicalization of enforcement as the gap between official prices and real prices widened. This mechanism is documented in granular detail for the period September 1793 to July 1794.

When Escobedo turns to the Essene communities of the first century BCE, the documentary base shrinks to Josephus, Philo of Alexandria, Pliny the Elder, and the Dead Sea Scrolls. The available evidence shows communal property, prohibition of commerce, ritualized distribution by a priestly elite, and expulsion of members who engaged in private trade. The structure is recognizable from the Jacobin case, but the mechanism cannot be reconstructed with the same granularity. Escobedo's implicit argument is uniformitarian: if the observable process in the data-rich case (France 1793-1794) produces a specific institutional pattern, and the fragmentary evidence from the data-poor case (Judean desert, first century BCE) is consistent with the same pattern, the extrapolation is warranted.

The strength of the extrapolation depends on the plausibility of the uniformity assumption. In Escobedo's case, the assumption is that human responses to commerce, inequality, and institutional intermediation are sufficiently stable across cultures and centuries to make extrapolation legitimate. The evidence he musters, spanning Greco-Roman antiquity, medieval Christendom, the Islamic world, early modern Europe, and the twentieth century, supports this assumption without proving it. Darwin faced the same limitation: the uniformity of natural processes was an assumption, not a theorem. But in both cases, the assumption becomes more credible as the number of independently documented instances increases.

B. Sequencing: Ordering Societies as Phases of a Recurring Cycle

Escobedo did not merely arrange his cases chronologically. He ordered them as phases of a recurring process, each illustrating a different stage in the anti-commercial cycle. Sparta represents the early phase of militarized anti-commercialism: the complete suppression of trade, the expulsion of foreigners, the subordination of individual autonomy to collective discipline. The Jacobins represent an intermediate phase: the state takes over the functions of commerce (price controls, requisitions, managed distribution) while maintaining the rhetoric of popular sovereignty. The

Soviet Union represents the terminal phase: total suppression of commercial activity, centralized planning, and the creation of a permanent bureaucratic class that administers distribution.

The societies are not presented as an actual lineage (the Jacobins did not descend biologically from the Spartans). They are presented as coexisting configurations that, when ordered, reveal the logic of a process that unfolds similarly wherever certain conditions obtain. This is Darwin's sequencing method translated to institutional analysis: just as Darwin used the flatworm eye, the nautilus eye, and the vertebrate eye as existing structures illustrating the logical possibility of gradual evolution, Escobedo used Sparta, Jacobin France, and the Soviet Union as existing historical configurations illustrating the logical structure of anti-commercial escalation.

The method yields additional analytical power when applied to intermediate cases. Escobedo's treatment of the Anabaptist commune of Munster (1534-1535) illustrates a transitional phase between Spartan militarism and Jacobin statism: Jan van Leiden established communal property and compulsory redistribution, but the enforcement mechanism was theocratic rather than bureaucratic. The medieval guilds represent another intermediate: they did not prohibit commerce outright but restricted it to licensed members, creating an intermediary class that captured rents from the exclusion of unlicensed competitors. Each intermediate case fills a gap in the sequence, just as each intermediate eye structure in Darwin's argument fills a gap between the simplest and most complex existing forms.

The sequencing method also enables what might be called a reverse reading: if the sequence describes stages of escalation, then the stages can be read backwards as stages of de-escalation. Escobedo does not develop this reverse reading systematically, but it is implicit in his treatment of periods where commercial order was restored after anti-commercial episodes. The post-Terror recovery in France, the post-Mao opening in China, the post-Soviet transition in Eastern Europe: each represents a reversal along the escalation sequence. The observation that reversals are slower and more difficult than escalations, that the path back is not the inverse of the path forward, is consistent with the hysteresis mechanism developed in Section IV.

C. Consilience: Accumulation Across Diverse Domains

This is Escobedo's central method and his greatest strength. The sheer range of his evidence is what makes the three volumes compelling. Ancient Greece, sectarian Judaism, early Christianity, medieval Islam, European heresies (Cathars, Waldensians, Anabaptists), the Reformation, the Counter-Reformation, Enlightenment rationalism, Romantic reaction, utopian socialism, Marxism, anarchism, fascism, Soviet communism, Maoism, Castroism: each domain independently exhibits the anti-commercial pattern. The explanation that each instance responds to its own local circumstances (political crisis, religious fervor, class resentment, imperial collapse) becomes less parsimonious than the explanation that a common replicative mechanism underlies the pattern.

The force of Escobedo's consilience depends on the independence of his domains, and here a qualification is necessary that distinguishes his case from Darwin's. Darwin's domains were causally independent: biogeography has no causal connection to embryology, the fossil record has no causal connection to the taxonomy of living species. The convergence of independent lines of evidence was what made the argument devastating. Escobedo's domains are partially connected genealogically. Marx read the Jacobins, the Jacobins read Rousseau, Rousseau idealized Sparta through Plutarch. There is direct memetic transmission across many of the cases.

This partial genealogical connection does not destroy Escobedo's consilience, but it modifies its character. Where the cases are genealogically connected (Marx to Proudhon to Rousseau to Sparta-via-Plutarch), what Escobedo documents is a chain of memetic transmission, not consilience of independent domains. Where the cases are genealogically disconnected (the Essenes and the Jacobins, the Inca communism and Maoism, the Anabaptist community of goods and Soviet collectivization), the convergence is genuine and the force of consilience applies. Escobedo's argument is strongest precisely in the pairs where no transmission chain connects the cases, because there the structural identity of the anti-commercial pattern must be explained by something other than cultural borrowing.

The distinction between connected and disconnected cases within Escobedo's corpus maps onto a distinction in evolutionary biology that Darwin himself recognized: homology (similarity due to common descent) versus analogy (similarity due to convergent evolution under similar selective pressures). Escobedo's connected cases are homologies: Marx's anti-commercialism resembles Rousseau's because it descends from it. Escobedo's disconnected cases are analogies: Inca collectivism resembles Soviet collectivism not because one descends from the other, but because similar environmental pressures (inequality, available intermediaries, weakened enforcement) selected for similar institutional responses.

A systematic classification of Escobedo's cases by the homology-analogy distinction would strengthen the consilience argument by identifying which convergences are genuinely independent. The following pairs illustrate the range:

Homologous pairs (connected by documented transmission): Marx and the Jacobins (Marx explicitly studied the French Revolution and drew lessons from it); Proudhon and Rousseau (Proudhon read Rousseau and built on his critique of property); the Anabaptists and the early Christian communities (the Anabaptists explicitly invoked the community of goods described in Acts 2:44-45); Lenin and Marx (direct intellectual descent, with Lenin's *What Is to Be Done?* building explicitly on Marxist theory).

Analogous pairs (no documented transmission): the Essenes and the Cathars (separated by twelve centuries and with no documented chain of transmission); Inca collectivism and Soviet collectivism (geographically and culturally disconnected); Spartan anti-commercialism and Japanese Tokugawa-era restrictions on merchants (independent developments in radically different cultural contexts); Chinese Legalism under the Qin dynasty and Jacobin statism (structurally similar but without genealogical connection).

The analogy pairs are the ones that carry the weight of Escohotado's consilience argument, because they cannot be explained by cultural borrowing and therefore require a structural explanation. The fact that geographically and temporally disconnected societies independently develop similar anti-commercial institutions under similar environmental conditions (inequality, intermediary availability, enforcement weakness) is powerful evidence for the replicative mechanism proposed here. The homology pairs, while less probative for the consilience argument, are valuable for a different reason: they document the transmission chain through which the anti-commercial meme propagates, providing evidence about the mechanism of memetic replication that the analogy pairs alone cannot supply.

D. The Missing Method: What Escohotado Did Not Do

Escohotado did not systematically employ discordance. He did not read institutional imperfections, vestigial norms, or regulatory structures that persist without serving their original function as evidence of institutional descent. His method was genealogical (tracing transmission chains) and consilient (accumulating evidence across domains), but not paleontological in the Darwinian sense of inferring history from vestiges.

This is not a criticism. Escohotado's project was historiographic, not mechanistic. He aimed to document the pattern, not to formalize the mechanism. But the absence of discordance from his methodological toolkit leaves a gap that becomes visible only when the toolkit is compared to Darwin's. Darwin used imperfections to establish the reality of descent with modification; Escohotado did not use institutional imperfections to establish the reality of memetic descent. ***The fourth method is precisely what Extended Phenotype Theory, operating through the concept of the institutional palimpsest and the mechanism of regulatory hysteresis, supplies.***

IV. THE FOURTH METHOD: INSTITUTIONAL DISCORDANCE AND THE LEGAL PALIMPEST

A. Dawkins's Palimpsest and Institutional Vestiges

In *The Genetic Book of the Dead*, Dawkins (2024) described the organism as a palimpsest: a manuscript in which earlier writing has been superimposed by later writing, but in which the earlier layers remain partially readable. The genome, Dawkins argued, is a record of ancestral environments. The human spine reads "quadruped" in bold lettering, overwritten too shallowly with "biped." Blood plasma reads "Paleozoic ocean." Embryonic gill arches read "aquatic ancestor." The palimpsest is not a metaphor; it is a specification of how descent with modification leaves traces that can be read by those who know what to look for.

Legal systems are palimpsests in precisely Dawkins's sense. Every constitution, code, and body of precedent carries layers of normative writing from successive institutional environments, partially overwritten but never fully erased. The Argentine Civil and Commercial Code, enacted in 2015,

replaced the 1871 Code drafted by Dalmacio Velez Sarsfield following the model of Napoleon's Code Civil of 1804. Yet the new Code preserves structural patterns inherited from the earlier one: the organization of property rights, the hierarchy of contractual obligations, the architecture of tort liability. These patterns persist not because they are optimal for the Argentine economy of 2026 but because the cost of complete institutional redesign exceeds the cost of toleration, just as the cost of redesigning the vertebrate retina (which is wired backwards, with the neural wiring in front of the photoreceptors) exceeds the cost of the slight optical inefficiency the design produces.

In prior work, I developed the concept of the institutional palimpsest in the context of memetic pincers and institutional field distortion (Lerer 2025b). The present analysis extends this concept by connecting it explicitly to Gould's analysis of Darwin's discordance method. The institutional palimpsest is not merely a convenient description of legal layering; it is the legal equivalent of the biological vestiges that Darwin used as evidence of descent. Just as fetal whale teeth reveal cetacean ancestry, vestigial norms reveal institutional ancestry.

B. Regulatory Hysteresis as the Mechanism of Vestige Production

The mechanism that produces institutional vestiges is regulatory hysteresis: the permanent deformation of institutional structures that persists after the originating norms are modified or repealed (Lerer 2025c). The concept is borrowed from physics, where hysteresis describes the phenomenon in which a system's output depends not only on its current input but on its history. A magnetized piece of iron does not return to its unmagnetized state when the external field is removed; some residual magnetization persists. The path back is not the inverse of the path forward.

In institutional terms, regulatory hysteresis means that repealing a norm does not eliminate the institutional structures that the norm generated during its period of operation. Argentina's collective bargaining agreements illustrate this: an agreement negotiated for the metallurgical industry in 1975 remains in force in 2025, its clauses governing working conditions that no longer exist in factories that have closed, for technologies that have been superseded, in economic conditions that bear no resemblance to those under which the agreement was drafted. The agreement is an institutional vestige in precisely Darwin's sense: a structure that persists not because it serves its current bearer but because it was inherited from an ancestor where it was functional.

The vestigial character of such norms becomes visible only when the analyst adopts the discordance method. A conventional legal analysis would ask: what function does this clause serve? If it serves no current function, it is an anomaly to be corrected by legislative reform. The discordance method asks a different question: from what institutional ancestor did this clause descend, and through what mechanism was it transmitted? The answer reveals not an anomaly but a trace of history, readable as evidence of the institutional environment that produced the clause, just as the human appendix reveals not a design flaw but a trace of a diet that included more cellulose than the modern human diet provides.

Additional examples abound. Argentina's rent control legislation of 2020 (Ley 27.551) generated a set of institutional responses (tenant organizations, specialized judicial chambers, administrative enforcement agencies) that persisted after the law was effectively repealed in late 2023. The organizations did not dissolve; they reoriented toward advocacy for the next round of controls. The judicial chambers did not close; they accumulated precedent that constrains future deregulation. These structures are institutional vestiges of a regulatory environment that no longer exists in its original form, and they generate the hysteretic resistance that makes the next reform more difficult than the last.

Escobedo documents analogous phenomena throughout his three volumes without naming the mechanism. The guilds of late medieval Europe, originally designed to ensure quality control and training in skilled trades, persisted for centuries after their quality-control function had been absorbed by other institutions. They survived not because they served the economy but because the institutional infrastructure they had built (political connections, legal privileges, training monopolies, residential concentration) generated self-reinforcing dynamics that resisted dissolution. When the French Revolution abolished the guilds in 1791 (the *Loi Le Chapelier*), the formal structures disappeared but the patterns of occupational restriction they had established persisted informally and were eventually reconstituted under new names. The guilds were vestigial institutions in the biological sense; their abolition did not erase the palimpsest layer they had inscribed.

The ultraactivity of collective agreements (the legal doctrine that an expired agreement continues in force until replaced by a new one) is the legal equivalent of what Gould and Lewontin (1979) called a spandrel: a structural byproduct of architectural constraints that may acquire secondary functions but was not selected for its current role. Ultraactivity was not designed to fossilize labor relations; it was designed to prevent gaps in worker protection during renegotiation. But the structural constraint it produces (the impossibility of returning to a pre-agreement baseline) generates vestigial norms that persist indefinitely, accumulating as layers in the institutional palimpsest.

Reading these vestiges as evidence of institutional descent, rather than as design failures to be corrected, is the methodological contribution of discordance applied to law. The vestigial collective agreement clause does not need to be explained by asking what function it currently serves (it may serve none). It needs to be explained by asking from what institutional ancestor it descended and through what mechanism it was transmitted. This is Darwin's fourth method, applied to institutional analysis.

C. Constitutional Lock-in as Accumulated Discordance

The Constitutional Lock-in Index (CLI), which I developed in prior work to measure institutional rigidity (Lerer 2025d), can be reinterpreted through the lens of discordance. A high CLI score indicates that the system carries a large number of institutional layers that resist modification: constitutional provisions, entrenched judicial doctrines, specialized courts, professional curricula,

union legal departments, and international treaties that reinforce the structure from above. Each of these layers is, in Dawkins's palimpsest terminology, a piece of writing from a previous institutional environment that has not been erased.

Argentina's CLI of 0.89 (on a scale of 0 to 1) for the labor regime means that the palimpsest is nearly illegible: so many layers of institutional writing have been superimposed that the original text is barely visible, and any attempt to modify the uppermost layer encounters resistance from the layers beneath. Chile's CLI of 0.24, by contrast, indicates a more legible palimpsest: fewer entrenched layers, greater institutional plasticity. The 23 failed labor reform attempts in Argentina over 34 years (1991 to 2024) are not 23 independent failures; they are 23 demonstrations that the accumulated layers of institutional writing resist erasure, just as 23 attempts to straighten the human spine would fail because the writing "quadruped" is inscribed too deeply in the vertebrate body plan.

The correlation between high CLI and zero sustained reform success (Argentina, CLI 0.89, 0% sustained success; Chile, CLI 0.24, multiple successful reforms) is the institutional equivalent of the correlation between vestigial complexity and evolutionary constraint. Systems with more vestiges are harder to modify because each vestige is connected to other structures that depend on it, just as the backwards-wired retina cannot be fixed without rewiring the entire visual processing system.

D. Pseudonorms: The Institutional Equivalent of Pseudogenes

Dawkins (2024) devoted particular attention to pseudogenes: genes that once performed useful functions but have been silenced by mutation, remaining in the genome as non-functional sequences that are "never read, never transcribed, never converted into proteins." The human genome is full of olfactory receptor pseudogenes, relics of an ancestral dependence on smell that modern humans have largely abandoned. The pseudogenes are informative precisely because they are non-functional: their presence reveals an ancestral environment where they were functional, and their continued existence reveals that the cost of removing them is lower than the cost of maintaining the genomic machinery that would be required for their removal.

Legal systems contain pseudonorms: statutes, regulations, and doctrinal principles that remain formally in force but are never applied, never cited in judicial decisions, never invoked by practitioners. They sit in the code like pseudogenes sit in the genome: readable to anyone who looks, but effectively silent. Argentina's Commercial Code of 1862, formally superseded by the unified Civil and Commercial Code of 2015, contained provisions on riverine commerce and horse-drawn transport that had been pseudonorms for decades before formal repeal. More significantly, portions of the labor regulatory framework contain provisions drafted for an industrial economy of the 1940s that have no application to the service economy of the 2020s but remain formally in force because the cost of identifying and repealing them individually exceeds the cost of their continued, silent presence.

The diagnostic value of pseudonorms lies in what they reveal about the institutional environment that produced them. A cluster of pseudonorms related to import-substitution industrialization reveals an ancestral regulatory environment oriented toward manufacturing autarky. A cluster of pseudonorms related to price controls reveals an ancestral inflationary environment. Reading pseudonorms is reading the institutional palimpsest at its deepest layers, where the earliest writing is most faded but most revealing. Dawkins noted that pseudogenes are "a lucid message from the past: a significant portion of the genetic book of the dead." Pseudonorms are a lucid message from the institutional past: a significant portion of the legal book of the dead.

This analysis connects directly to Escobar's empirical program. Many of the anti-commercial institutions he documented have left pseudonorm traces in legal systems that have since liberalized. Spain's post-Franco transition, for example, dismantled the corporatist economic framework of the Franco era, but vestigial provisions of the *Fuero del Trabajo* (1938) persisted in Spanish labor law for decades after the transition to democracy. These pseudonorms, unreadable without the discordance method, reveal the ancestral anti-commercial environment that produced them and provide diagnostic information about the points of institutional vulnerability where the anti-commercial meme might reactivate if environmental conditions shift.

V. THE GENEALOGICAL TRUNK: SMITH, DARWIN, DAWKINS, HAYEK, AND ESCOBAR

The methodological convergence between Darwin and Escobar is not a coincidence to be noted and set aside. It reflects a deeper genealogical unity in the intellectual traditions that produced both arguments.

A. Adam Smith as Common Intellectual Ancestor

In *The Wealth of Nations* (1776), Adam Smith articulated the foundational insight that complex functional order can emerge from the uncoordinated actions of individuals pursuing their own interests, without any designer or central planner. The division of labor, the price system, the emergence of money as a medium of exchange: all are instances of order without design. Smith's "invisible hand" is the economic equivalent of what Darwin would later call natural selection: a mechanism that produces adaptive order as a byproduct of individual competition, not as an intended outcome.

Darwin read Thomas Malthus, who built directly on Smith's logic of decentralized competition under scarcity. In *An Essay on the Principle of Population* (1798), Malthus extended Smith's analysis of economic competition to biological populations. Darwin's notebook entry of September 28, 1838, records the moment of insight: reading Malthus provided the mechanism that connected variation and inheritance to differential survival. The lineage runs Smith to Malthus to Darwin.

Gould (2002) recognized this genealogical connection and drew attention to its radicalism. Darwin, Gould argued, took Smith's paradox (that the most effective route to an orderly economy

is not central planning but individual competition) and transferred it intact to the natural world. In Smith's framework, organisms assume the role of competing firms, and reproductive success becomes the equivalent of profit. The "invisible hand" that produces economic order without intention becomes natural selection producing biological order without foresight. Gould emphasized that this was Darwin's most subversive move against Paley's natural theology: he accepted Paley's empirical observation that organisms exhibit exquisite design, but demonstrated that the design was a side effect of a blind, amoral, individualistic struggle for reproductive success. The harmony of nature, like the efficiency of the market, was a byproduct of competition, not the intention of a designer.

On the other branch, Carl Menger (1871) developed the theory of spontaneous institutional emergence (most notably in his account of the origin of money) in explicit dialogue with the Scottish Enlightenment tradition that Smith had founded. Ludwig von Mises (1949) and Friedrich Hayek (1973, 1988) extended Menger's program into a complete theory of spontaneous order. Hayek was explicit about the evolutionary character of his analysis: institutions emerge from human action but not from human design, through a process of cultural group selection that operates analogously (though not identically) to biological natural selection.

The two lineages converge in Richard Dawkins. *The Selfish Gene* (1976) and *The Extended Phenotype* (1982) completed the Darwinian program by specifying the gene as the unit of selection and extending the concept of phenotype beyond the organism's body to include artifacts that genes build in the world to enhance their own replication. The beaver dam, the spider web, the caddisfly case: all are extended phenotypes, products of gene action that modify the environment in ways that favor the replicating gene. Hayek's "extended order" of human cooperation, the set of institutions (property rights, contract law, money, markets) that emerge spontaneously from individual action and enable large-scale anonymous cooperation, is, I have argued elsewhere (Lerer 2025a), the memetic equivalent of Dawkins's extended phenotype. The parallel terminology is not coincidental; it reflects structural identity.

Escotado sits on the Hayekian branch, though he does not cite Hayek extensively. His intellectual sympathy is with the tradition of classical liberalism: Smith, the Scottish Enlightenment, and the defense of commerce as a civilizing force. His empirical project, documenting the recurrence and persistence of anti-commercial ideology, is intelligible only within a framework that treats commerce as the baseline of social order and anti-commercialism as a deviation requiring explanation. That framework is Smith's, transmitted through the Austrian tradition. Escotado's genealogy of the enemies of commerce is, in this sense, the empirical complement to Hayek's theoretical analysis of the extended order: where Hayek explained how spontaneous commercial order emerges, Escotado documented how it is attacked, suppressed, and, sometimes, destroyed.

B. The Reunification, Not Eclecticism

The integration of Extended Phenotype Theory with Escobar's empirical program is therefore not an eclectic combination of unrelated traditions. It is the reunification of two branches that diverged from a common trunk. Smith's insight about order without design was developed by Darwin into the theory of natural selection and by Menger into the theory of spontaneous institutional emergence. Dawkins refined Darwin's program by specifying the replicator (gene) and extending the phenotype beyond the body. Hayek refined Menger's program by extending the concept of spontaneous order to encompass the entire institutional framework of civilization.

Hayek was remarkably explicit about the parallel. In *The Fatal Conceit* (1988), he wrote that the extended order resulted not from human design or intention but from spontaneously conforming to certain traditional and largely moral practices. The phrase "extended order" is terminologically parallel to "extended phenotype," and the conceptual structure is identical: both describe complex functional structures that emerge from the interaction of individual units (genes, actors) without central design. Hayek could not specify the replicator that generated his extended order because the concept of the meme had not yet been formalized when he began his theoretical program (Dawkins introduced the term in 1976; Hayek's key works on spontaneous order date from 1960 and 1973). The EPT framework completes Hayek's program by specifying memes as the replicators whose competition generates the institutional structures that constitute the extended order.

Conversely, the Hayekian framework completes Dawkins's program in the institutional domain by providing what *The Extended Phenotype* lacked: a theory of how memetic phenotypes interact to produce large-scale institutional order. Dawkins's examples of extended phenotypes in the biological domain (beaver dams, spider webs, caddisfly cases) are single-replicator constructions. The institutional extended phenotype is a multi-replicator construction: property law is not the extended phenotype of a single meme but the emergent result of competition and cooperation among thousands of memes operating through the actions of millions of individuals. Hayek's theory of spontaneous order provides the framework for understanding how individual-level memetic competition aggregates into institutional-level patterns without central coordination.

Escobar sits -one more time- on the Hayekian branch, documenting what happens when the spontaneous order is attacked.

C. The Anti-Commercial Intermediary as Niche Constructor

The reunification illuminates a phenomenon that neither branch fully explains alone: the role of the anti-commercial intermediary as a niche constructor. In biological terms, niche construction is the process by which organisms modify their environment in ways that alter the selective pressures acting on themselves and on other organisms. The beaver builds a dam that creates a pond that changes the ecosystem, which in turn affects the selective pressures on the beaver and on every other species in the altered environment.

The anti-commercial intermediary, in Escohotado's account, is a niche constructor. The feudal lord who suppresses commercial activity in his territory is not merely responding to an environment where commerce is absent; he is constructing the environment where commerce is absent, and the absence of commerce creates the conditions that make his intermediation necessary. The union leader who blocks labor market reform is not merely operating in a rigid labor market; he is constructing the rigidity, and the rigidity creates the conditions that make his representation indispensable. The clientelistic distributor of social transfers is not merely serving a population that lacks access to markets; he is maintaining the conditions under which market access is impossible, and the impossibility creates the demand for his distributive services.

This niche construction operates through the extended phenotype. The intermediary's memes construct institutional phenotypes (specialized courts, licensing requirements, price controls, mandatory intermediation) that modify the environment in ways that favor the replication of the same memes. The feedback loop is self-reinforcing: the memes build institutions, the institutions modify the environment, the modified environment selects for the memes. The result is what Escohotado documented across 2,600 years: the anti-commercial pattern that persists not because it produces good outcomes for the societies that adopt it, but because the intermediaries who propagate it construct the niche that ensures their own survival.

Escohotado's empirical work thus documents what happens when the spontaneous order is attacked by niche-constructing intermediaries. The EPT framework provides the mechanism that explains why the attacks recur. The two branches reunite in the present analysis, where Escohotado's three methods (uniformity, sequencing, consilience) are complemented by the fourth method (discordance) that the EPT framework provides. The reunification is possible because both branches descend from Smith, and the same logic of order-without-design that makes commercial cooperation intelligible also makes the persistence of anti-commercial ideology intelligible: both are emergent phenomena driven by the differential replication of informational units (genes, memes) competing for representation in the next generation.

VI. FROM DESCRIPTION TO MECHANISM: WHAT THE INTEGRATED FRAMEWORK PREDICTS

Escohotado described the pattern with extraordinary empirical richness. What he did not provide, because his project was genealogical rather than formal, is a mechanistic explanation for the pattern's recurrence and a set of falsifiable predictions derived from that explanation.

The integrated framework, combining Escohotado's consilience with EPT's discordance method, generates three predictions that Escohotado's genealogy alone cannot produce:

A. Prediction 1: The Three-Condition Activation Threshold

If the anti-commercial meme replicates through the mechanism proposed here, rather than through ideological transmission alone, then its activation should require three simultaneous conditions:

visible inequality (the emotional fuel that makes the meme psychologically attractive), organized intermediaries who benefit from the suppression of anonymous cooperation (the transmission vehicle), and weakened enforcement of the institutional framework that protects commercial exchange (the window of opportunity). Each condition is necessary; none is sufficient alone.

This prediction is falsifiable. It would be refuted by a major anti-commercial episode occurring in the absence of any one of the three conditions: an anti-commercial movement succeeding in a society with low inequality and no intermediary class, or a society with high inequality and organized intermediaries but strong enforcement nevertheless experiencing anti-commercial capture. Escobar's own data can be used to test the prediction retrospectively: the three conditions should be present in every case he documents and absent (or incomplete) in every society that resisted anti-commercial capture during the same period.

Argentina in the period 2020-2023 constitutes a contemporary test. The three conditions were simultaneously present: a Gini coefficient of 0.43 (INDEC), 21 million beneficiaries of social transfer programs organized through clientelistic distribution networks, and collapse of enforcement (illegal property occupations without criminal prosecution, contractual obligations without effective judicial remedy). The anti-property meme flourished with particular virulence during this period, consistent with the prediction.

B. Prediction 2: Vestigial Norm Density Correlates with Reform Failure

If institutional vestiges accumulate as layers in a palimpsest and generate resistance to modification through regulatory hysteresis, then the density of vestigial norms (norms that persist without serving their original function) should correlate with reform failure rates, independently of the technical quality of the proposed reform. A well-designed reform should fail at the same rate as a poorly designed reform in a system with high vestigial density, because the resistance comes from the accumulated layers, not from the content of the proposal.

This prediction is testable using the CLI framework and Argentina's 23 reform attempts. If the prediction holds, the technical quality of each reform proposal (measured independently by legal scholars) should show no correlation with success or failure. What should correlate with failure is the number of entrenched institutional layers the reform would have to modify: specialized courts, existing jurisprudence, professional training programs, union legal departments, and international treaty obligations.

Preliminary evidence is consistent. Argentina's most technically sophisticated reform attempt, the Milei government's 182-article Labor Modernization bill (TCI 0.79), faces the same institutional resistance as cruder earlier attempts. The variable that predicts failure is not design quality but palimpsest depth.

C. Prediction 3: Discordant Institutions Identify the Next Vulnerability

If institutional vestiges are readable as evidence of descent (the fourth method), then the location and character of vestigial norms should identify the points of greatest vulnerability to the next anti-commercial episode. Specifically, the sectors where vestigial density is highest should be the sectors where intermediaries can most easily recreate the conditions for anti-commercial capture, because the institutional infrastructure for intermediation already exists as a vestige even if the original anti-commercial regime has been partially reformed.

This prediction has practical implications for institutional design. If regulators could read the institutional palimpsest, identifying vestigial structures the way a biologist identifies vestigial organs, they could anticipate where the next anti-commercial cycle is most likely to emerge and direct preventive resources accordingly. The prediction is falsifiable: the next anti-commercial episode should emerge in the sector with the highest vestigial density, not in the sector with the highest current inequality or the most organized intermediaries.

Argentina's labor sector provides a test case. The vestiges are dense: 187 specialized labor courts, 3,847 union legal departments, mandatory labor law curricula in all 109 law schools, 185 bar associations with labor law sections, and international treaties with constitutional rank (ILO Conventions 87 and 98, ICESCR) that reinforce the structure from above. If the prediction holds, the next anti-commercial episode in Argentina should manifest first in the labor sector, even if inequality is more visible in the housing sector or intermediaries are more organized in the social welfare sector. The vestigial infrastructure determines the point of entry, not the current distribution of grievances.

By contrast, sectors where vestiges have been cleared (Argentina's telecommunications sector, substantially deregulated in the 1990s with few vestigial structures remaining) should be resistant to anti-commercial capture even under favorable environmental conditions, because the institutional infrastructure for intermediation does not exist as a prebuilt platform. The anti-commercial meme would need to construct its extended phenotype from scratch in such sectors, a process that is slower and more visible than the reactivation of existing vestigial structures.

VII. DISCUSSION: IMPLICATIONS FOR INSTITUTIONAL THEORY

The analysis has several implications that extend beyond the Escobar-Darwin parallel.

First, it suggests that the methodological tools developed for evolutionary biology are not merely analogically useful for institutional analysis but formally applicable. Gould's four methods of historical inference were designed for a specific domain (the reconstruction of biological history from present evidence), but their logical structure is domain-independent. Any discipline that seeks to reconstruct past processes from present evidence faces the same inferential challenges and can benefit from the same methodological toolkit. Escobar's independent reinvention of three of the four methods suggests that the toolkit is naturally discovered by careful practitioners of historical inference, whether or not they are aware of its biological origins.

Second, the identification of discordance as the missing method in Escobedo's program has implications for the broader project of evolutionary institutional analysis. Institutional economists in the tradition of Douglass North (1990) and Daron Acemoglu and James Robinson (2012) have documented institutional persistence and path dependence extensively. But the dominant metaphor is lock-in: institutions persist because switching costs are high. The palimpsest metaphor adds something that lock-in does not: the idea that institutions carry readable traces of their history that can be used diagnostically. Lock-in tells you that reform is difficult; palimpsest analysis tells you why, by identifying the specific layers of institutional writing that generate resistance.

Third, the genealogical trunk connecting Smith, Darwin, Menger, Dawkins, Hayek, and Escobedo suggests that the fragmentation of institutional analysis across disciplinary boundaries (evolutionary biology, economics, law, political science, history) may be concealing a unity that practitioners in each discipline would benefit from recognizing. Escobedo, working from the history of ideas, and Dawkins, working from evolutionary biology, arrived at convergent conclusions about the persistence of maladaptive informational patterns. The convergence is explicable once the common intellectual ancestor (Smith) is identified and the logic of order-without-design is recognized as the shared foundation.

Fourth, the framework contributes to the Gould-Dawkins debate itself. Gould's (2002) multilevel selection theory, with its emphasis on species selection and macroevolutionary contingency, might seem to favor approaches that treat institutions as units of selection in their own right. The present analysis suggests a different integration: Gould's methodological contribution (the four methods of historical inference) is fully compatible with Dawkins's gene-centered (or in the institutional case, meme-centered) framework. The methods are tools for reading evidence; the theoretical framework determines what the evidence means. Discordance as a method is neutral between gene-centered and species-selection accounts; it is the interpretation of vestiges as products of memetic competition, rather than as failures of collective design, that connects the method to the EPT framework.

Fifth, the analysis illuminates a point about Escobedo's reception that has puzzled commentators. Despite the extraordinary erudition and documentary range of *Los enemigos del comercio*, the trilogy has not achieved the academic influence that its quality would seem to warrant. The framework developed here suggests a partial explanation: Escobedo's work lacks a formal mechanism. Academics trained in the social sciences expect not merely pattern documentation but causal explanation. Escobedo's "sempiternidad del mensaje" (the everlasting quality of the message) is a description, not a mechanism. By supplying the mechanism (memetic replication through extended phenotype construction, operating under the three-condition activation threshold and generating hysteretic vestiges that accumulate in the institutional palimpsest), the EPT framework may make Escobedo's empirical contribution more legible to audiences that require formalization as a condition of engagement.

Sixth, the distinction between homologous and analogous cases within Escobedo's corpus has methodological implications for comparative institutional analysis more broadly. When legal

scholars compare institutions across jurisdictions, they rarely distinguish between similarity due to common descent (legal transplants, shared colonial heritage, diffusion through international organizations) and similarity due to convergent evolution under similar selective pressures. The homology-analogy distinction from evolutionary biology provides a principled basis for making this distinction. Two jurisdictions that adopted similar anti-commercial regulations because one copied the other (homology) present a different analytical situation from two jurisdictions that independently developed similar regulations in response to similar environmental pressures (analogy). The policy implications differ: homologous similarity can be undone by tracing and interrupting the transmission chain; analogous similarity requires modifying the environmental pressures that select for the convergent outcome.

VIII. LIMITATIONS

Several limitations constrain the present analysis and should be noted explicitly.

First, the analogy between biological and institutional vestiges is structural, not ontological. Biological vestiges are products of genetic descent with modification; institutional vestiges are products of memetic transmission with hysteretic deformation. The mechanisms differ in important respects: genetic transmission has high fidelity; memetic transmission does not. Biological evolution is blind; institutional change involves intentional actors. These differences do not invalidate the structural analogy, but they constrain the precision with which biological models can be applied to institutional phenomena.

Second, the assessment of Escobar's methods is retrospective and interpretive. Escobar did not describe his methodology in the terms used here; the classification of his methods as instances of Gould's four types is my own analytical contribution. It is possible that Escobar would have rejected the classification, or that a different analyst would categorize his methods differently. The argument stands or falls on whether the structural isomorphism between Darwin's methods and Escobar's is genuine, not on whether Escobar intended it.

Third, the partial genealogical connection among Escobar's cases (the fact that many anti-commercial movements borrowed directly from earlier ones) weakens the consilience argument relative to Darwin's, where the independence of domains was more complete. The analysis acknowledges this limitation and proposes the homology/analogy distinction as a way of addressing it, but a fully rigorous assessment would require coding each of Escobar's cases for the presence or absence of genealogical connections to other cases and testing whether the pattern holds among the genuinely independent ones.

Fourth, the falsifiable predictions in Section VI are formulated at a level of generality that makes testing difficult. The three-condition activation threshold -in particular- requires operational definitions of "visible inequality," "organized intermediaries," and "weakened enforcement" that may admit of reasonable disagreement. Future work should specify threshold values and coding protocols that would permit systematic testing.

IX. CONCLUSION

Darwin described *The Origin of Species* as one long argument. Escohotado produced three long volumes. The structural parallel is not superficial: both projects deployed accumulative evidential strategies to establish patterns that the prevailing intellectual orthodoxy of their respective times was predisposed to reject. Darwin faced creationism; Escohotado faced the presumption that anti-commercial movements are rational responses to injustice rather than instances of a replicative phenomenon.

Of the four methods Gould identified in Darwin's argumentative strategy, Escohotado employed three: uniformity, sequencing, and consilience. The fourth, discordance, the reading of imperfections as vestiges of historical descent, is what Extended Phenotype Theory adds. The institutional palimpsest (Dawkins 2024), regulatory hysteresis, and the Constitutional Lock-in Index provide the tools for reading institutional vestiges the way Darwin read rudimentary organs: not as design failures but as traces of descent that reveal the history of the system.

The methodological convergence is not accidental. It reflects the genealogical unity of two intellectual traditions, biological evolutionism and spontaneous order theory, both of which descend from Adam Smith's foundational insight that complex functional order can emerge without a designer. Escohotado sits on the Hayekian branch, documenting what happens when the spontaneous order is attacked. The EPT framework sits on the Dawkins branch, providing the mechanism that explains why the attacks recur. The reunion of the two branches produces an integrated framework with diagnostic and predictive power that neither branch possesses alone.

If the framework is correct, three predictions follow. The anti-commercial cycle activates only when three conditions are simultaneously present: visible inequality, organized intermediaries, and weakened enforcement. Vestigial norm density predicts reform failure independently of reform quality. And the sectors with the highest vestigial density are the sectors most vulnerable to the next anti-commercial episode.

Escohotado spent the last decade of his life on the trilogy. His work deserves readers who engage it seriously with new tools, not to demonstrate that he was wrong (on the essentials, he was right), but to explore why the patterns he documented recur with the regularity they exhibit. If the evolutionary mechanism proves illuminating, the credit belongs to Escohotado for having identified the pattern with sufficient precision that a formal framework could be coupled to it. If it adds nothing, the problem lies with the framework, not with Escohotado.

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